

Question 1

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The functions f and g are defined by the given equations below, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $f(x) = 18(1.4)^{x+5}$

II. $g(x) = 18(1.16)(1.4)^{x+3}$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

In the xy -plane, the graph of function f , where $y = f(x)$, has exactly 8 x -intercepts. One of these x -intercepts is $(17, 0)$. The rational function g is defined by $g(x) = \frac{f(x)}{x - 17}$. In the xy -plane, how many x -intercepts does the graph of $y = g(x)$ have?

Answer Preview:

Question 3

3

Mark for Review



$$g(x) = x^3 + ax^2 + bx + c$$

The function g is defined by the given equation, where a , b , and c are integer constants. The zeros of the function are -2 , -7 , and 6 . What is the value of a ?

(A) -84

(B) -3

(C) 3

(D) 84

Question 4

$$f(x) = 3(x - a)(x - b)(x - c)$$

The function f is defined by the given equation, where a , b , and c are distinct constants. When $a < x < b$, the value of $f(x)$ is positive. The graph of $y = f(x)$ in the xy -plane contains the point (r, s) , where r and s are constants. If $s = 8$, which of the following could be true?

I. $r < a$

II. $b < r < c$

III. $r > c$

(A) I only

(B) III only

(C) I and III only

(D) II and III only

The function f is defined by $f(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $f(15a)$ and $f(8a)$?

Answer Preview:

Question 6

$$\frac{4}{9}(4x + 9)(x + \sqrt{4k + 9})(x - \sqrt{4k + 9}) = 0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 81. What is the value of k ?

Answer Preview:

Question 7

The function g is defined as $g(x) = \frac{2x - 4}{(x + 11)(x - 6)}$. If $g(a + 5) = 0$, where a is a constant, what is the value of a ?

Answer Preview:

The function f is defined by $f(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $f(15a)$ and $f(8a)$?

Answer Preview:

The function p is defined by $p(x) = a((x + 6)^2 - b)((x + 6)^2 - c)$, where a , b , and c are constants. In the xy -plane, the graph of $y = p(x)$ passes through the points $(-7, 24)$ and $(0, 899)$. What is the value of $p(-12) + p(-5)$?

Answer Preview:

In the xy -plane, the graph of function f , where $y = f(x)$, has exactly 8 x -intercepts. One of these x -intercepts is $(17, 0)$. The rational function g is defined by $g(x) = \frac{f(x)}{x - 17}$. In the xy -plane, how many x -intercepts does the graph of $y = g(x)$ have?

Answer Preview:

The function f is defined by $f(x) = 30x$. For what value of x is $f(x) = 180$?

Answer Preview:

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Question 12

For the polynomial function v , $v(2) = 3$ and $v(0) = -13$. The graph of $y = v(x)$ in the xy -plane passes through the points $(-2, 7)$ and $(6, 0)$. Which of the following must be true?

(A) When $v(x)$ is divided by $(x + 13)$, the remainder is 0

(B) When $v(x)$ is divided by $(x + 7)$, the remainder is -2.

(C) When $v(x)$ is divided by $(x + 2)$, the remainder is 3.

(D) When $v(x)$ is divided by $(x - 6)$, the remainder is 0.

Question 13

$$56(x^3 + 64)(x^4 - 81) = 0$$

How many distinct real solutions does the given equation have?

(A) Exactly two

(B) Exactly three

(C) Exactly five

(D) Exactly seven

The function f is defined by $f(x) = \frac{|x|}{a} - 16$, where $a < 0$. What is the product of $f(17a)$ and $f(8a)$?

Answer Preview:

Question 15

The function f is defined by $f(x) = -39^x$. The function g is a decreasing linear function. In the xy -plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points, (h, j) and (k, m) , where $j > m$. When $g(x) < f(x)$, which of the following must also be true?

(A) $x > k$

(B) $x < h$

(C) $x > k$ or $x < h$

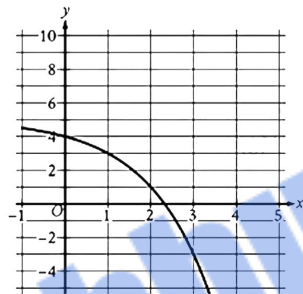
(D) $h < x < k$

$$7x^2(4x - 13)(4x - u) = 0$$

In the given equation, u is a positive constant. The sum of the solutions to the equation is 13. What is the value of u ?

Answer Preview:

Question 17



The graph of $y = f(x) + 2$ is shown. Which equation defines function f ?

(A) $f(x) = -2^x + 1$

(B) $f(x) = -2^x + 3$

(C) $f(x) = -2^x + 4$

(D) $f(x) = -2^x + 5$

Question 18

The function h is defined by $h(x) = 14(2)^{-x} + 7$. What is the value of $h(3)$?

(A) $\frac{42}{4}$

(B) $\frac{35}{4}$

(C) -77

(D) -105

Question 19

The functions g and h are defined by the given equations.

$$g(x) = \sqrt{(x-11)^2 + 45}$$

$$h(x) = \sqrt[3]{64} + \frac{x^2}{2}$$

If $g(8) = t$, where t is a constant, what is the value of $h(t)$?

(A) $3\sqrt{6}$

(B) 22

(C) 31

(D) 51

Question 20

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Mark for Review



For a polynomial function f , the graph of $y = f(x)$ in the xy -plane contains the points $(-6, 0)$, $(7, 0)$, $(0, 0)$, and $(4, 0)$. Which of the following must be a factor of $f(x)$?

(A) $x^2 - 2x - 24$

(B) $x^2 - x + 42$

(C) $x^2 - 11x - 28$

(D) $x^2 - 7x$

$$\frac{5}{8}(5x + 8)(x + \sqrt{5k + 8})(x - \sqrt{5k + 8}) = 0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 64. What is the value of k ?

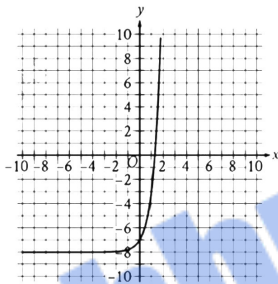
Answer Preview:

$$x(9x - 2a) + b(9x - 2a) - (x + b) = 0$$

In the given equation, a and b are positive constants. A solution to the equation is $x = 42$. What is the value of a ?

Answer Preview:

Question 23



The graph of the equation $y = 5^x + k$ is shown, where k is a constant. What is the value of k ?

(A) -8

(B) -7

(C) 7

(D) 8

Question 24

The function f is defined by $f(x) = -23^x$. The function g is a decreasing linear function. In the xy -plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points, (h, j) and (k, m) , where $j > m$. When $g(x) < f(x)$, which of the following must also be true?

(A) $x > k$

(B) $x < h$

(C) $x > k$ or $x < h$

(D) $h < x < k$